

Proof. The disjoint atoms in $(s, t]$ give

$$F_\xi(t) - F_\xi(s) = P\{s < \xi \leq t\} = \sum_{s < x \leq t} P\{\xi = x\}.$$

Thus, the increment is zero if there is no atom in $(s, t]$, and

$$F_\xi(x_i) - F_\xi(x_i-) = P\{\xi = x_i\}.$$

□

Definition. The joint distribution of

$$\xi_1, \xi_2, \dots, \xi_n$$

is a probability measure defined by

$$P_{\xi_1, \xi_2, \dots, \xi_n}(B) = P\{(\xi_1, \xi_2, \dots, \xi_n) \in B\}, \quad B \in \mathcal{B}(\mathbb{R}^n).$$

Definition. $\xi \in L^1(\Omega, \mathcal{F}, P)$ if

$$\int_{\Omega} |\xi| dP < \infty.$$

Then

$$E(\xi) = \int_{\Omega} \xi dP$$

exists and is called the expectation of ξ .

Example of computing expectation.

1. For $A \in \mathcal{F}$, $E(\mathbb{1}_A) = P(A)$.
2. If η is a step function, that is,

$$\eta = \sum_{i=1}^n a_i \mathbb{1}_{A_i}$$

with pairwise disjoint $A_i \in \mathcal{F}$, then

$$E(\eta) = \sum_{i=1}^n a_i P(A_i).$$

Exercise 1.7 (Law of the Unconscious Statistician(LOTUS)). If $h : \mathbb{R} \rightarrow \mathbb{R}$ is a Borel measurable function and $h(\xi)$ is integrable, then

$$E(h(\xi)) = \int_{\mathbb{R}} h(x) dP_\xi(x).$$

Hence,

1. $E(h(\xi)) = \int_{\mathbb{R}} h(x) dP_\xi(x)$ for a continuous law
2. $E(h(\xi)) = \sum_{i=1}^{\infty} h(x_i) P\{\xi = x_i\}$ for a discrete law.

Definition (Variance). $\xi \in L^2(\Omega, \mathcal{F}, P) \subset L^1$ if

$$E(|\xi|^2) = \int_{\Omega} |\xi|^2 dP < \infty.$$

Its variance is

$$\text{var}(\xi) = E((\xi - E(\xi))^2) = E(\xi^2) - (E(\xi))^2.$$

Exercice 1.8 (Cauchy-Schwarz inequality).

$$|E(\xi\eta)|^2 \leq E(\xi^2)E(\eta^2).$$

Consequently, $L^2 \subset L^1$, and

$$E(|\xi|) \leq \sqrt{E(\xi^2)}.$$

Proof. Let $\xi, \eta \in L^2$. Then

$$\xi\eta \in L^1 \quad \text{since} \quad 2|\xi\eta| \leq \xi^2 + \eta^2.$$

The quadratic function

$$t \mapsto E((\xi - t\eta)^2) = E(\xi^2) - 2tE(\xi\eta) + t^2E(\eta^2) \quad \text{is non negative.}$$

So its discriminant is non positive:

$$E(\xi\eta)^2 - E(\xi^2)E(\eta^2) \leq 0 \implies E(\xi\eta)^2 \leq E(\xi^2)E(\eta^2).$$

Apply this to ξ and $\mathbb{1}$ to

$$E(|\xi|) \leq \sqrt{E(\xi^2)}.$$

□

Exercice 1.9 (Coarea formula). If $\eta \geq 0$, then

$$E(\eta^2) = 2 \int_0^\infty tP\{\eta > t\}dt.$$

Proof. We have¹

$$\eta^2 = \int_0^\infty 2t\mathbb{1}_{\{\eta > t\}}dt.$$

Tonelli's theorem gives

$$\begin{aligned} E(\eta^2) &= \int_\Omega \int_0^\infty 2t\mathbb{1}_{\{\eta > t\}}dt dP \\ &= \int_0^\infty 2tE(\mathbb{1}_{\{\eta > t\}})dt \\ &= \int_0^\infty 2tP\{\eta > t\}dt. \end{aligned}$$

□

1.3 Conditional Probability and Independence

Definition. Let $A, B \in \mathcal{F}$ be such that $P(B) > 0$. The conditional probability of A given B is defined by

$$P(A|B) = \frac{P(A \cap B)}{P(B)}.$$

Exercice 1.10. If $B_1, \dots$ are pairwise disjoint, $\bigcup_{n \geq 1} B_n = \Omega$, and $P(B_n) > 0$, then

$$P(A) = \sum_{n \geq 1} P(A|B_n)P(B_n).$$

¹Note that $\int_0^\eta 2t\mathbb{1}_{\{\eta > t\}}dt \rightarrow \eta^2$ and $\int_\eta^\infty 2t\mathbb{1}_{\{\eta > t\}}dt \rightarrow 0$ since $\mathbb{1}_{\{\eta > t\}}$ is 1 when $\eta > t$ and 0 otherwise.

Proof. Since $A = \bigcup_{n \geq 1} (A \cap B_n)$, we have

$$P(A) = \sum_{n \geq 1} P(A \cap B_n) = \sum_{n \geq 1} P(A|B_n)P(B_n).$$

□

Remark (Bayes' rule). If $P(A) > 0$, then

$$P(B_j|A) = \frac{P(A|B_j)P(B_j)}{\sum_{n \geq 1} P(A|B_n)P(B_n)}.$$

Definition. Events A and B are independent if

$$P(A \cap B) = P(A)P(B).$$

Events $A_1, \dots, A_n$ are independent if for every subfamily,

$$P(A_{i_1} \cap \dots \cap A_{i_k}) = P(A_{i_1}) \dots P(A_{i_k}).$$

Exercise 1.11. If $P(B) > 0$, then

$$A \text{ and } B \text{ are independent} \iff P(A|B) = P(A).$$

Definition. Random variables η, ξ are independent if

$$P(\eta \in A, \xi \in B) = P(\eta \in A)P(\xi \in B)$$

for all $A, B \in \mathcal{B}(\mathbb{R})$. An arbitrary family is independent if every finite subfamily is independent.

Proposition 1.1. If $\xi_1, \dots, \xi_n$ are independent, then

$$E(\xi_1 \dots \xi_n) = \prod_{i=1}^n E(\xi_i).$$

In particular, independent $\xi, \eta \in L^2$ are uncorrelated:

$$E(\eta\xi) = E(\eta)E(\xi)$$